Project: Safe Hydra- Fire Hydrant Pump House Monitoring System

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Revision 1.0

Customer M/S Aragen Pharma, Nacharam

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Revision History

Version	Previous Ver	Date	Change Content
1.0	AA	3-Apr-6	Initial Release

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1. Project Overview

The fire pump house at Aragen Pharma, Nacharam plays a mission-critical role in plant safety, comprising a jockey pump, main hydrant pump, and diesel pump that must remain fully operational and ready at all times. Any unnoticed failure, pressure drop, or delayed response can significantly compromise fire preparedness. Conventional fire pump systems rely heavily on local panels, pressure switches, and manual checks, offering limited visibility into real-time pump health, operational status, or fault conditions especially during non-working hours.

To overcome these limitations, Eltrive Automations introduces SAFE HYDRA NextGen Hydrant Monitoring & Alert System, an advanced IoT-enabled solution designed to provide 24×7 real-time monitoring, instant alerts, and centralized visibility of fire pump operations. SAFE HYDRA enables early fault detection, proactive maintenance, faster response during emergencies, and improved system reliability, ensuring that the fire hydrant infrastructure remains always ready, compliant, and dependable.

SAFE HYDRA – Smart, proactive, and always-ready fire safety monitoring.

1.1 Benefits of SAFE HYDRA

1.1.1 24×7 real-time monitoring of jockey, main, and diesel fire pumps
1.1.2 Instant alerts for pump faults, pressure drops, and power failures
1.1.3 Proactive maintenance through early fault detection and trend analysis
1.1.4 Centralized and remote access to fire pump status from anywhere
1.1.5 Improved safety, compliance readiness, and system reliability

1.2 Key Features

1.2.1 Jockey pump health monitoring
1.2.2 Main hydrant pump health monitoring
1.2.3 Diesel pump health monitoring
1.2.4 Pressure, level, voltage, kw, kwh, and current sensing
1.2.5 Web and mobile dashboards with SMS / Email alerts
1.2.6 Customizable, compliance-ready reports

2. Scope of Work – SAFE HYDRA

2.1 Eltrive Scope

2.1.1 Site visit to the fire pump house and associated areas
2.1.2 Capturing and finalizing system requirements based on site conditions
2.1.3 Design and engineering of the SAFE HYDRA monitoring system
2.1.4 Procurement and supply of sensors, IoT controller, gateway, and accessories
2.1.5 Installation and integration of SAFE HYDRA electrical control panel
2.1.6 PLC programming, logic development, and testing
2.1.7 IoT configuration, dashboard setup, and alert logic implementation
2.1.8 Configuration of control panel signals to relay modules
2.1.9 Configuration of IoT module and network connectivity
2.1.10 End-to-end system testing and commissioning
2.1.11 Conduct training sessions for customer personnel
2.1.12 Provide system manuals and technical documentation
2.1.13 Complete system health checks for pumps, sensors, and dashboards

2.2 Customer Scope / Support Required

2.2.1 Installation of energy meters and CTs inside existing electrical panels using supplied materials
2.2.2 Electrical panel wiring, terminations, and modifications inside pump house panels
2.2.3 Execution support for fabrication and welding required for mounting sensors on diesel tanks and pipelines
2.2.4 Provision of hot work permissions for welding and mechanical activities
2.2.5 All statutory approvals, permits, and site clearances required during installation
2.2.6 Coordination for shutdowns, if required, during installation and commissioning
2.2.7 Ensuring a safe working environment and compliance with site safety norms
2.2.8 One dedicated technician or maintenance personnel support during installation
2.2.9 Civil works, supports, and structural modifications, if required
2.2.10 Execution support for cable laying, routing, and physical installation of supplied cables and trays
2.2.11 Incoming power supply and signal connections from existing panels.

2.3 Assumptions

2.3.1 Accessible pump room
2.3.2 Clean 230V AC power supply
2.3.3 Adequate space for equipment
2.3.4 Internet connectivity available
2.3.5 Customer to provide layout diagrams
2.3.6 Site access coordination by customer
2.3.7 Necessary approvals handled by customer
2.3.8 Any deviation from the above assumptions may impact project cost, schedule, or scope and shall be addressed through mutual discussion and formal change approval.

2.4 Out of Scope

2.4.1 Remote start/stop of pumps
2.4.2 Diesel pump auto-start
2.4.3 Major panel modifications
2.4.4 NFPA/IS compliance testing
2.4.5 Firefighting performance of pumps
2.4.6 Piping or civil modifications
2.4.7 Installation of pumps/valves/tanks
2.4.8 Extended AMC beyond warranty

3. Deliverables

3.1 Hardware Deliverables (BOQ)

S.No	Details	Qty	Unit	Price	Amount
1	**Full Hardware Kit** Including all sensors (Ultrasonic & Laser), PLC with complete control panel (including PLC unit, SMPS Delta/Schneider make), along with all required electrical components and mechanical brackets/mounting accessories.	1	Lot	5,29,811	**5,29,811**

3.2 Services Supply

S.No	Details	Make	Qty	Amount
16	PLC Configuration & Programming	Eltrive	1	20000
17	Web Application Development and Customization charges	Eltrive	1	80000
18	Erection and Commissioning	Eltrive	1	40000
19	Subscription- Yearly (Server and Data)	Eltrive	1	1,25,000
	Service Supply Total			**2,65,000**

3.3 Software Charges

20	Software Application Development charges • Customised as per company specific themes and colours • Customised test sequences & Health conditions • Customised logbooks & Reports • Configuration of SMS & Email Alerts	**1,45,000**

3.4 Optional Services

Two Free users already included for additional user charges as below

21	Android Application subscription per user per year	20000
22	IOS application subscription per user per year	35000

4. User Access & Application Availability

4.1 Mobile Application Access

4.1.1 A dedicated mobile application will be provided for real-time alerts and notifications. The application will be pre-configured to enable automatic access without requiring repeated logins.

4.1.2 One (1) mobile user license is included free of cost under the base package. Any additional mobile users will be charged as per the applicable commercial pricing table.

4.2 Web Application Access

4.2.1 Web-based dashboard access will be provided through secure login credentials. The base package includes complimentary access for:

- Two (2) Admin Users
- Three (3) Standard Users

4.2.2 Admin Users will have configuration, monitoring, and system management rights.

4.2.3 Standard Users will have viewing and operational access based on assigned permissions.

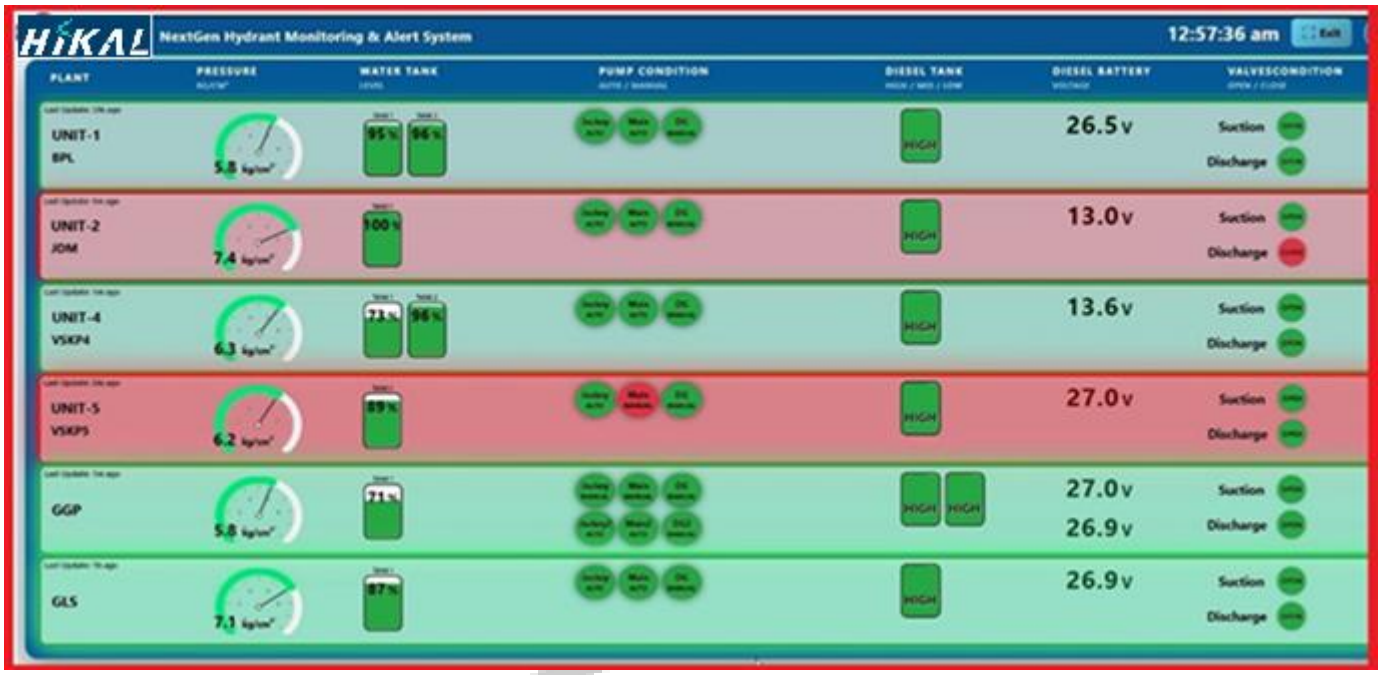
4.3 Additional Users

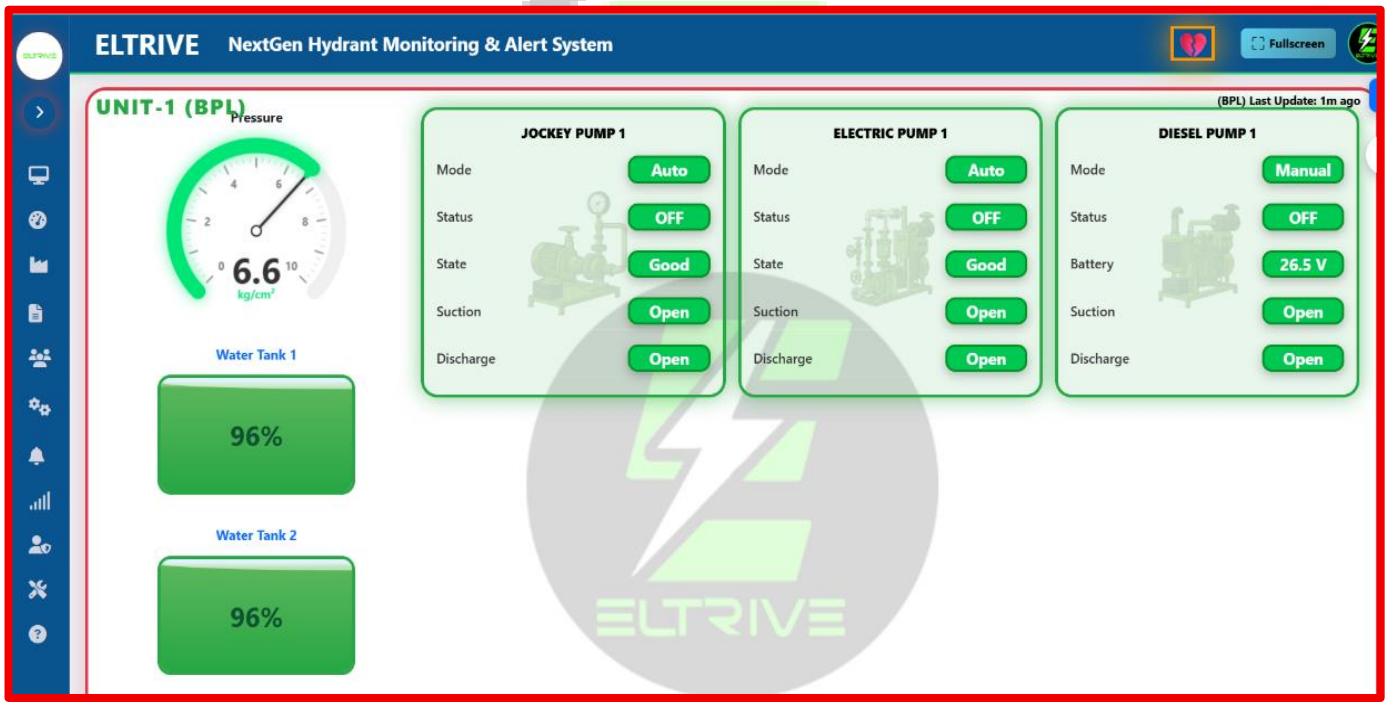
4.3.1 Any requirement for additional mobile or web user accounts beyond the above-mentioned limits can be provisioned at an additional commercial cost, in accordance with SafeHydra licensing terms and pricing.

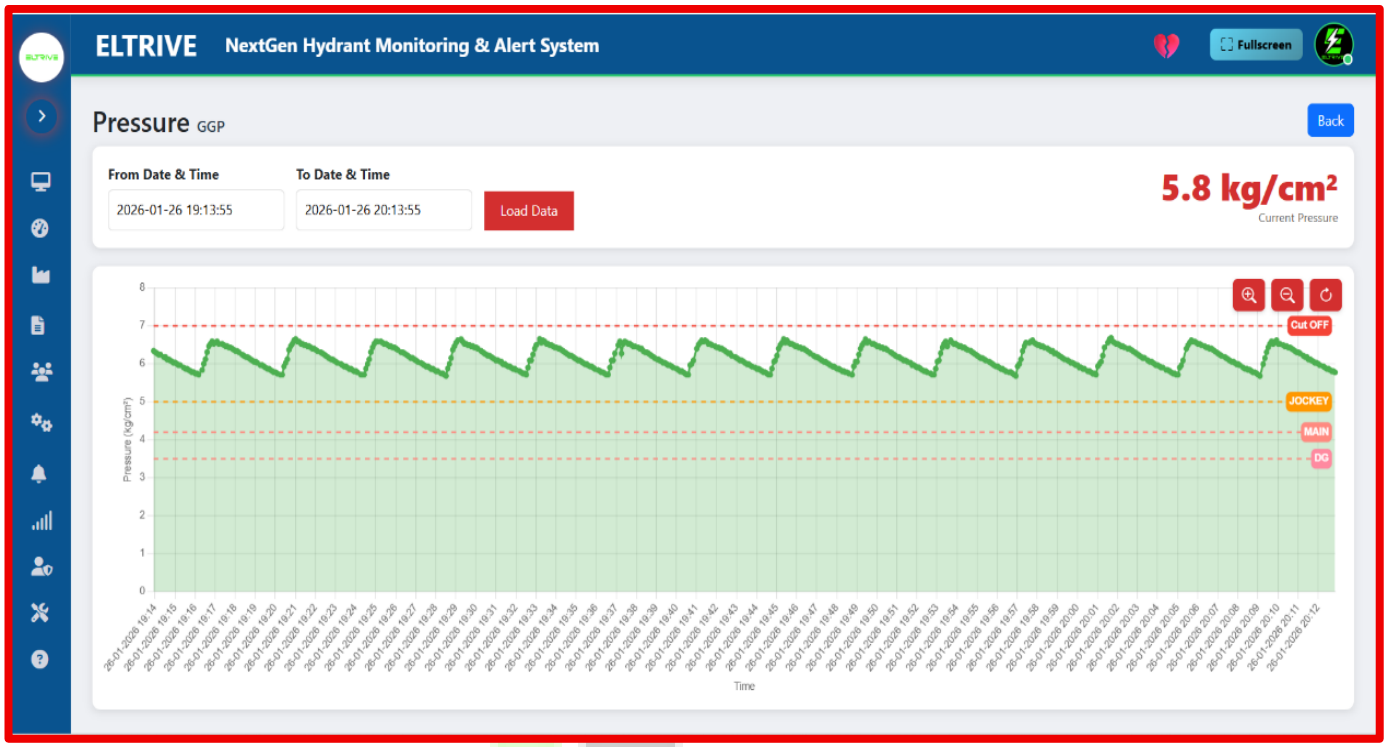
4.4 Access Control & Security

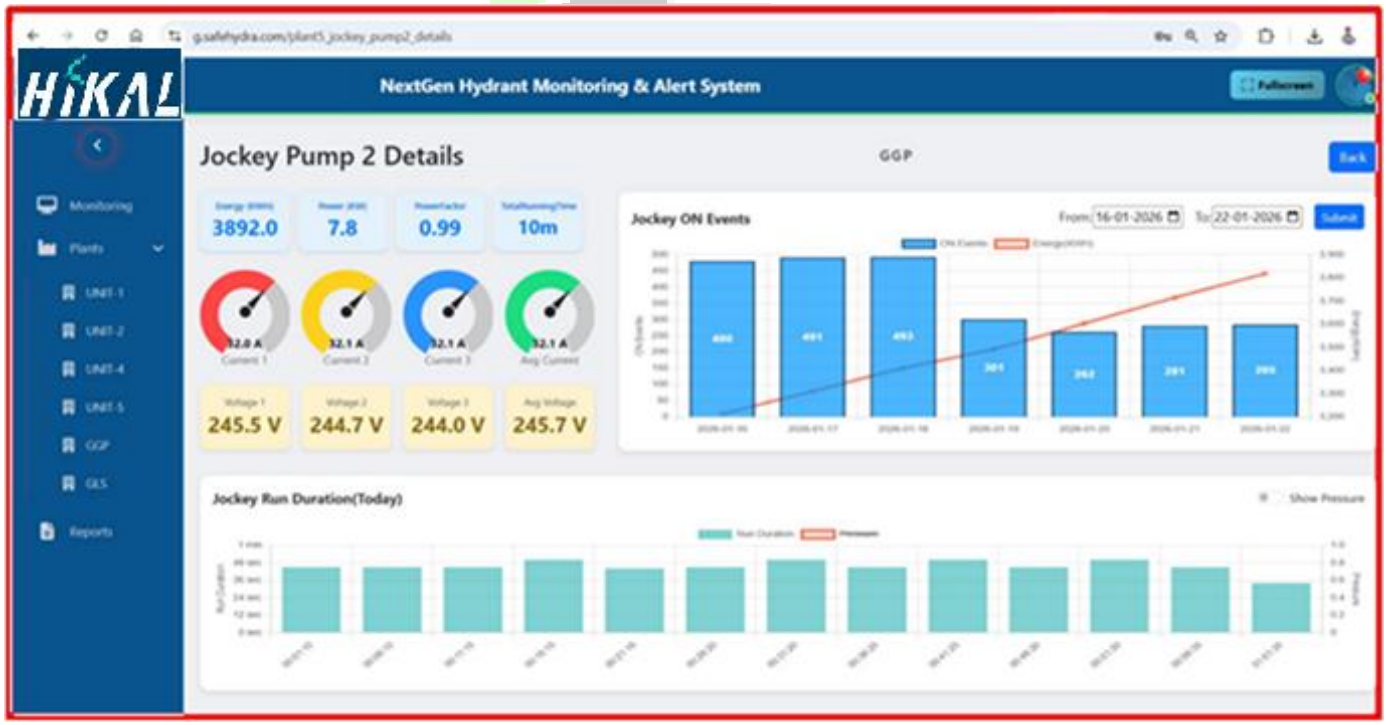
4.4.1 All user accounts will be role-based and password-protected to ensure secure, controlled, and traceable access to system data, alerts, and reports, supporting compliance with pharmaceutical industry data integrity and security standards.

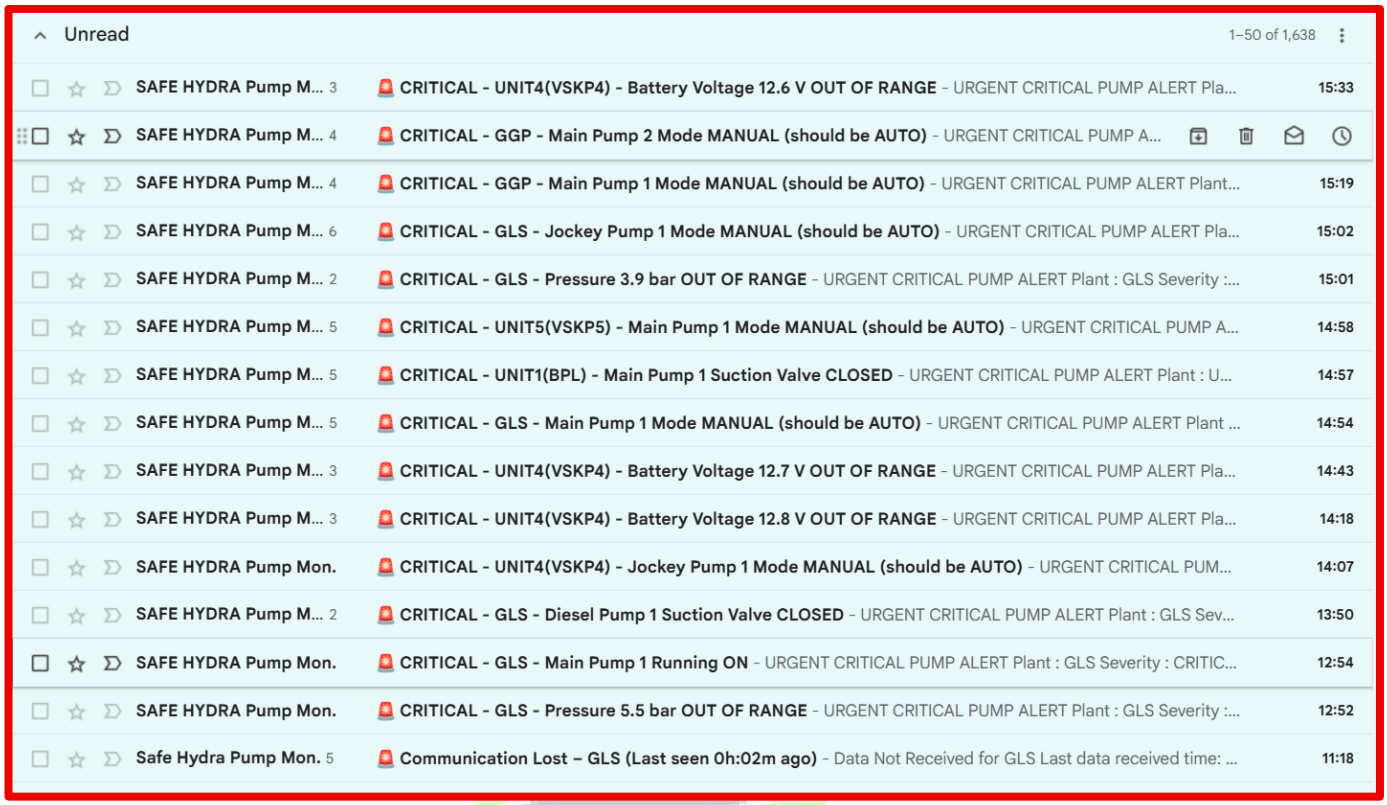
5. Application Reference Images





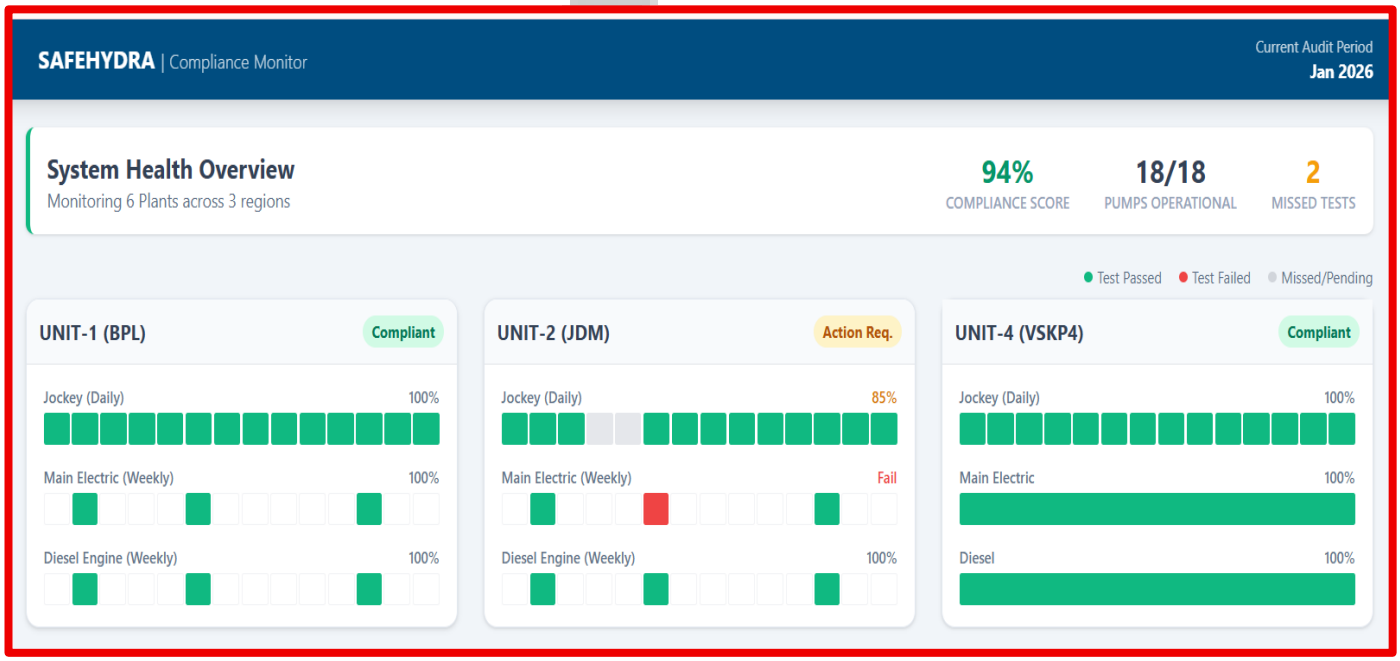

ELTRIVE
NextGen Hydrant Monitoring & Alert System
Fullscreen
Pressure GGP
Back
From Date & Time
2026-01-26 19:13:55
To Date & Time
2026-01-26 20:13:55
Load Data
5.8 kg/cm²
Current Pressure
Cut OFF
JOCKEY
MAIN
DG
Pressure (kg/cm²)
Time


HIKAL
NextGen Hydrant Monitoring & Alert System
Fullscreen
Jockey Pump 2 Details
GGP
Back
Monitoring
Plants
UNIT-1
UNIT-2
UNIT-4
UNIT-5
GGP
GLS
Reports
3892.0
7.8
0.99
10m
245.5 V
244.7 V
244.0 V
245.7 V
Jockey ON Events
From 16-01-2026
To 22-01-2026
Submit
Jockey Run Duration(Today)
Show Pressure

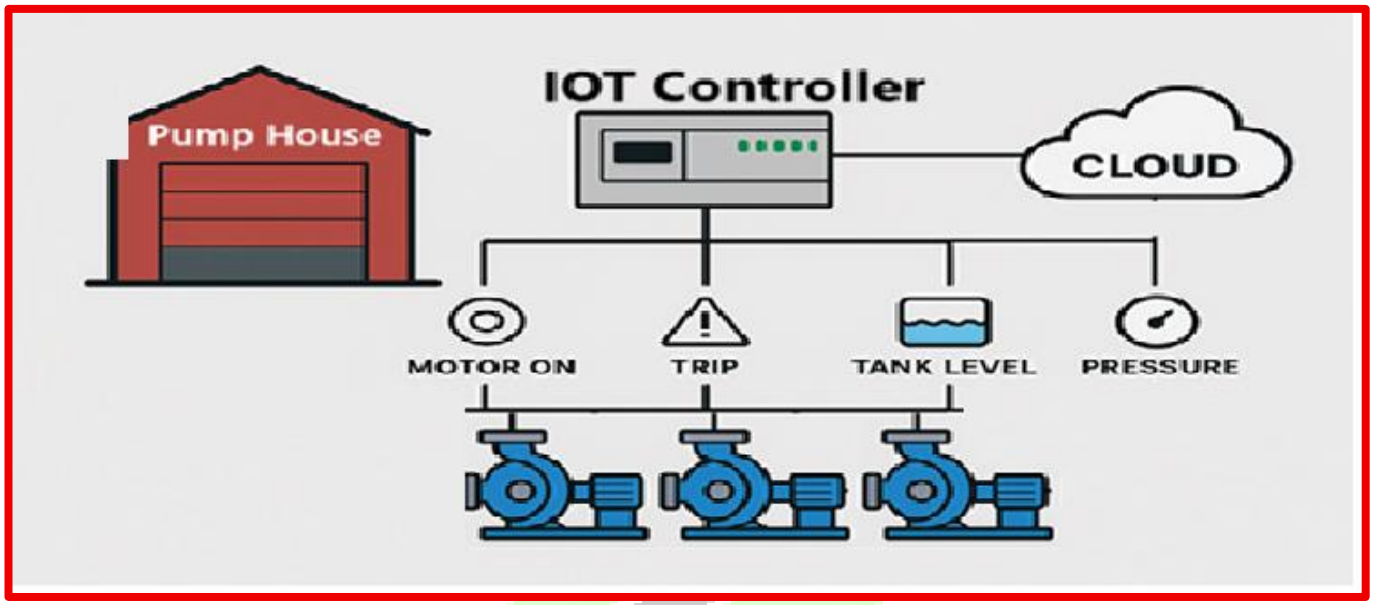


Unread 1–50 of 1,638

SAFE HYDRA Pump M... 3	CRITICAL - UNIT4(VSKP4) - Battery Voltage 12.6 V OUT OF RANGE - URGENT CRITICAL PUMP ALERT Pla...	15:33
SAFE HYDRA Pump M... 4	CRITICAL - GGP - Main Pump 2 Mode MANUAL (should be AUTO) - URGENT CRITICAL PUMP A...	
SAFE HYDRA Pump M... 4	CRITICAL - GGP - Main Pump 1 Mode MANUAL (should be AUTO) - URGENT CRITICAL PUMP ALERT Plant...	15:19
SAFE HYDRA Pump M... 6	CRITICAL - GLS - Jockey Pump 1 Mode MANUAL (should be AUTO) - URGENT CRITICAL PUMP ALERT Pla...	15:02
SAFE HYDRA Pump M... 2	CRITICAL - GLS - Pressure 3.9 bar OUT OF RANGE - URGENT CRITICAL PUMP ALERT Plant : GLS Severity :...	15:01
SAFE HYDRA Pump M... 5	CRITICAL - UNIT5(VSKP5) - Main Pump 1 Mode MANUAL (should be AUTO) - URGENT CRITICAL PUMP A...	14:58
SAFE HYDRA Pump M... 5	CRITICAL - UNIT1(BPL) - Main Pump 1 Suction Valve CLOSED - URGENT CRITICAL PUMP ALERT Plant : U...	14:57
SAFE HYDRA Pump M... 5	CRITICAL - GLS - Main Pump 1 Mode MANUAL (should be AUTO) - URGENT CRITICAL PUMP ALERT Plant ...	14:54
SAFE HYDRA Pump M... 3	CRITICAL - UNIT4(VSKP4) - Battery Voltage 12.7 V OUT OF RANGE - URGENT CRITICAL PUMP ALERT Pla...	14:43
SAFE HYDRA Pump M... 3	CRITICAL - UNIT4(VSKP4) - Battery Voltage 12.8 V OUT OF RANGE - URGENT CRITICAL PUMP ALERT Pla...	14:18
SAFE HYDRA Pump Mon.	CRITICAL - UNIT4(VSKP4) - Jockey Pump 1 Mode MANUAL (should be AUTO) - URGENT CRITICAL PUM...	14:07
SAFE HYDRA Pump M... 2	CRITICAL - GLS - Diesel Pump 1 Suction Valve CLOSED - URGENT CRITICAL PUMP ALERT Plant : GLS Sev...	13:50
SAFE HYDRA Pump Mon.	CRITICAL - GLS - Main Pump 1 Running ON - URGENT CRITICAL PUMP ALERT Plant : GLS Severity : CRITIC...	12:54
SAFE HYDRA Pump Mon.	CRITICAL - GLS - Pressure 5.5 bar OUT OF RANGE - URGENT CRITICAL PUMP ALERT Plant : GLS Severity :...	12:52
Safe Hydra Pump Mon. 5	Communication Lost – GLS (Last seen 0h:02m ago) - Data Not Received for GLS Last data received time: ...	11:18



6. Site wise Proposals



7. Commercials

7.1 Price Breakup

Item	Details	HSN	Qty	Unit	Amount(Rs)
1	Hardware Supply	85371	1	Lot	5,29,811
2	Service & with AMC	998314	1	Lot	2,65,000
4	Software Applications	998314	1	Lot	1,45,000
				Total Amount	**9,39,811**

7.2 Tension Free AMC

7.2.1 Yearly AMC Charges – **2,50,000**

7.2.1.1 Subscription Cost yearly 1,25,000 (First year already covered)

7.2.1.2 Maintenance & Service of HW 1,25,000

7.2.1.3 Including 4 visits

7.2.1.4 Additional visits 15000 / Day

7.2.1.5 First year all included.

7.3 Total Cost

7.3.1 Total Cost for the first year including subscription Rs.**9,39,811**

7.4 Payment Terms

7.4.1 Validity: This quotation is valid for 30 days from the issue date.

7.4.2 Taxes: GST Extra 18%

7.4.3 Installation & Commissioning: **Included.**

7.4.4 Material Delivery Cost: **Included.**

7.4.5 Warranty 1 year from the delivery date

7.4.6 Delivery: 2 weeks from the date of the PO.

7.4.7 Payment Terms: 50% advance, 50% against delivery